DTU Wind and Energy Systems

Phase-based optimization of wind turbine placement in offshore wind farms

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DTU Wind-B-0167

May 2026

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ECTS: 15

Education: Bachelor of Science

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Remarks:
This report is submitted as partial fulfillment of the requirements for graduation in the above education at the Technical University of Denmark.

DTU Wind and Energy Systems is a department of the Technical University of Denmark with a unique integration of research, education, innovation and public/private sector consulting in the field of wind and energy. Our activities develop new opportunities and technology for the global and Danish exploitation of wind energy. Research focuses on key technical-scientific fields, which are central for the development, innovation and use of wind and energy and provides the basis for advanced education.

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Before reading

Table of contents

List of Tables

List of Figures

1 Introduction

1.1 Motivation

One of the biggest challenges in the world today is greenhouse gas emissions and global warming. In Denmark, the government has set a target to reduce emissions by 82% by 2035 [1]. One of the main goals is to quadruple the production of solar energy and onshore wind energy, while also significantly increasing offshore wind energy by 2030 [2]. The current situation is more critical than ever before, but studies show that there is still a long way to go, and Denmark has only reached 7% of the 2030 goals for onshore wind energy production. Denmark's large sea areas and strong wind resources make it highly suitable for offshore wind energy production, particularly in the North Sea. As part of the Danish energy agreements, a minimum of 4 GW of new offshore wind capacity is planned to be tendered before 2030. In addition, the Danish Finance Act of 2022 includes plans to tender a further 2 GW of offshore wind capacity within the same timeframe. These initiatives reflect Denmark's ambition to significantly expand its offshore wind sector as part of the green transition. While wind turbine technology continues to develop, the rate of performance improvements has slowed compared to earlier decades. As a result, recent advancements in offshore wind primarily focus on scaling up wind farm size, increasing turbine capacity, and optimizing wind farm layouts to reduce wake losses and maximize AEP. Consequently, increasing attention is now directed towards improving the overall efficiency of wind farms by not only utilizing larger turbines, but also by optimizing their spatial arrangement. This enables a more efficient use of the available area, ensuring that energy production is maximized within the spatial constraints of the wind farm.

1.2 Project description

This Bachelor's thesis focuses on the optimization of offshore wind farm layouts. The study investigates several optimization methods and algorithms commonly used within wind farm layout optimization, including a phase-based greedy heuristic (SmartStart and 2-step), a gradient-based optimization approach, and a stochastic Random Search method. The phase-based greedy algorithm constructs and refines layouts sequentially by placing and relocating turbines based on incremental improvements in Annual Energy Production (AEP). The gradient-based method, implemented through TOPFARM, performs continuous optimization by adjusting turbine positions using gradient information to improve the layout locally. In contrast, the Random Search method explores the design space through stochastic perturbations of turbine positions, providing diverse initial layouts for further optimization. The objective of this thesis is to provide a systematic and analytical comparison of these methods in terms of both precision and computational efficiency. The analysis is conducted using the TOPFARM framework [3] in combination with the PyWake simulation tool [**pywake2.5.0˙2023**], which enables accurate evaluation of AEP while accounting for wake effects.

The study is guided by the following research questions:

- How does the choice of initial layout affect the performance of wind farm layout optimization?
- How effective is a phase-based greedy algorithm for wind farm layout optimization?

- How effective is a gradient-based algorithm for optimization for a wind farm?
- How do the different methods compare, and which is the most effective?

2 Litereture review

Wind farm layout optimization has been extensively studied, primaiyly with the objective of maximizing the annual energy production (AEP), by minimizing wake losses between turbines. The problem is complex, where many variables takes part, the power output of each turbine depends not only on the local wind conditions but also on the positions of all other turbines. As a result, the layout optimization problem is highly non-linear and computationally expensive, especially for large wind farms. Early approaches to wind farm layout design were based on simple engineering guidelines and regular grid patterns, where turbines were spaced according to empirical rules of thumb. However, with the increasing size and complexity of modern wind farms, these approaches have largely been replaced by computational optimization methods. These methods aim to systematically explore the design space to identify layouts that maximize AEP under given constraints.

A key challenge in wind farm optimization is the trade-off between model accuracy and computational efficiency. High-fidelity wake models, such as analytical or Gaussian-based models, which provides a more accurate predictions of wake effects but are computationally expensive. Simpler AEP calculations and Wake models, on the other hand, allow for faster evaluations but may reduce accuracy. This trade-off is highlighted by Bueno [4], who demonstrates that computationally efficient AEP models can significantly reduce runtime while still providing sufficiently accurate results for optimization purposes. This suggests that faster evaluation methods can enable more extensive exploration of the design space, which may ultimately lead to better overall solutions.

In addition to model selection, the choice of optimization algorithm plays a crucial role. Greedy algorithms represent one of the simplest approaches to wind farm layout optimization. These methods iteratively place turbines by selecting the position that provides the largest immediate increase in AEP. Chen et al. [5] show that greedy algorithms can produce competitive layouts but with a cost of higher computational cost. However, due to their sequential and locally optimal nature, they are prone to converging to local optima and may fail to identify globally optimal configurations. To address these limitations, more advanced optimization techniques have been developed. One method is Gradient-based algortihm, which have gained increasing attention due to their computational efficiency and scalability. These methods uses the gradients of the layout to efficiently navigate the solution and can significantly reduce the number of required function evaluations. Valotta Rodrigues et al. [6] demonstrate that gradient-based optimization can outperform traditional heuristic methods in terms of computational speed, especially for large-scale wind farms. Furthermore, their results highlight the importance of a good initial layout, as gradient-based methods are sensitive to the starting point and may otherwise converge to suboptimal solutions.

Another important aspect of wind farm optimization is the formulation of the problem itself. In many studies, the layout optimization problem is treated as a continuous optimization problem, where turbine positions can vary freely within the site boundaries. However, in practical applications, the problem is often discretized into a finite set of candidate positions, leading to a combinatorial selection problem. Acheimastos [7] investigates this discrete formulation and compares different optimization methods for selecting turbine positions. The study finds that gradient-based methods offer high computational efficiency, while greedy approaches can achieve slightly higher AEP in some cases. Importantly, the results also show that down-selection methods can achieve near-optimal performance, with differences in AEP of less than 0.15% compared to fully optimized layouts. This finding is particularly relevant for real-world wind farm development, where layouts are often designed in stages. In practice, the exact turbine specifications may not be known at the initial design stage, and layouts may need to be adapted later as new turbine technologies become available. Despite this, most existing studies consider wind farm layout optimization as a single-stage problem with fixed turbine characteristics.

Recent work has also emphasized the importance of computational efficiency in enabling large-scale optimization. Sarcos et al. [8] highlight that reducing computational cost is essential for exploring a larger number of design configurations, particularly when dealing with many turbines and wind conditions. Their work demonstrates that improvements in algorithm design and model efficiency can significantly accelerate the optimization process, making it feasible to handle more complex and realistic scenarios.

Overall, the literature shows that no single optimization method is universally superior. Greedy algorithms are simple and fast but may lead to suboptimal solutions, while gradient-based methods are efficient and scalable but depend heavily on the initial layout. Hybrid approaches that combine heuristic initialization with gradient-based refinement appear to offer the best performance in many cases. However, a notable gap in the literature is the limited focus on sequential optimization problems, where an initial layout is first generated and later refined or reduced. While Acheimastos [7] demonstrates the potential of down-selection, there is still a lack of systematic investigation into multi-stage optimization strategies that combine different methods across different stages of the design process. This gap motivates the approach taken in this project, where a combination of heuristic and gradient-based methods is used to improve wind farm layouts with a focus on maximizing AEP.

3 Theory

3.1 Weibull Wind Speed Distribution

When planning a wind farm, one of the most important factors is the wind resource. Wind velocity is typically measured over long time periods, often spanning at least one year. The Weibull distribution is widely used to describe the statistical distribution of wind speeds in terms of both frequency and intensity. The Weibull distribution can be expressed either as a probability density function (PDF) or a cumulative distribution function (CDF). The PDF is given by:

$$f(u) = \frac{k}{A}\left(\frac{u}{A}\right)^{k-1} \exp\left(-\left(\frac{u}{A}\right)^{k}\right) \tag{1}$$

where k is the shape parameter and A is the scale parameter. In practice, wind speed data is often discretized into bins, where each bin represents a range of wind speeds, for example intervals of $1\,m/s$.

While the Weibull distribution describes the overall wind speed distribution at a site, wind conditions also vary with direction. To account for this, separate Weibull distributions can be defined for different wind sectors. This allows both the magnitude and frequency of wind to be described as a function of direction, providing a more complete characterization of the wind climate.

3.2 Wind Turbines and Power Extraction

The performance of a wind turbine is governed by its ability to extract kinetic energy from the wind. A typical modern wind turbine is a horizontal-axis, three-bladed machine defined by key parameters such as rotor diameter, hub height, and rated power. The power extracted from the wind can be expressed as

$$P = \frac{1}{2} A \rho C_p V^3 \tag{2}$$

where A is the rotor swept area, ρ is the air density, V is the wind speed, and C_p is the power coefficient. The power coefficient represents the efficiency of the turbine and is bounded by the Betz limit, which states that no more than 59.3% of the kinetic energy in the wind can be converted into mechanical energy.

3.3 Annual Energy Production (AEP)

Annual Energy Production (AEP) is a central performance metric used to evaluate the expected energy output of a wind farm over one year [9]. Since wind conditions vary in both wind speed and wind direction, the produced power must be weighted by the probability of each wind state occurring. In practice, the continuous probability distributions of wind speed and direction are discretized into a finite number of bins. The wind direction is divided into N_θ sectors, and the wind speed into N_u bins. This discretization enables a computationally efficient approximation of the expected energy production. The AEP can then be approximated as

$$AEP \approx 8760 \sum_{d=1}^{N_\theta} \sum_{u=1}^{N_u} \sum_{n=1}^{N_{WT}} P_n(u_{\text{eff},n}(\theta_d, u_u)) \cdot \rho_{d,u}, \tag{3}$$

where $P_n(u_{\text{eff},n})$ is the power produced by turbine n at the effective wind speed $u_{\text{eff},n}$, which accounts for wake effects. The term $\rho_{d,u}$ represents the joint probability of wind speed and direction, and 8760 corresponds to the number of hours in a year. This formulation corresponds to a numerical approximation of the expected value of power production over the wind climate [10].

3.4 Wake Effects in Wind Farms

When a wind turbine extracts energy from the wind, it creates a wake region characterized by reduced wind speeds and increased turbulence [9]. These wake effects influence downstream turbines, which experience lower inflow velocities and therefore produce less power. Since the power output of a turbine depends on the cube of the wind speed, even small reductions in velocity can result in significant energy losses. In addition, increased turbulence leads to higher structural loading and fatigue. Wake effects are therefore a key factor in wind farm design and must be considered when estimating energy production ansd optimizing layouts [11].

3.4.1 N.O. Jensen Wake Model

One of the earliest and most widely used wake models is the Jensen model [12]. It provides a simplified analytical description of the wake using a uniform velocity deficit (top-hat profile). The wake radius is assumed to expand linearly with downstream distance:

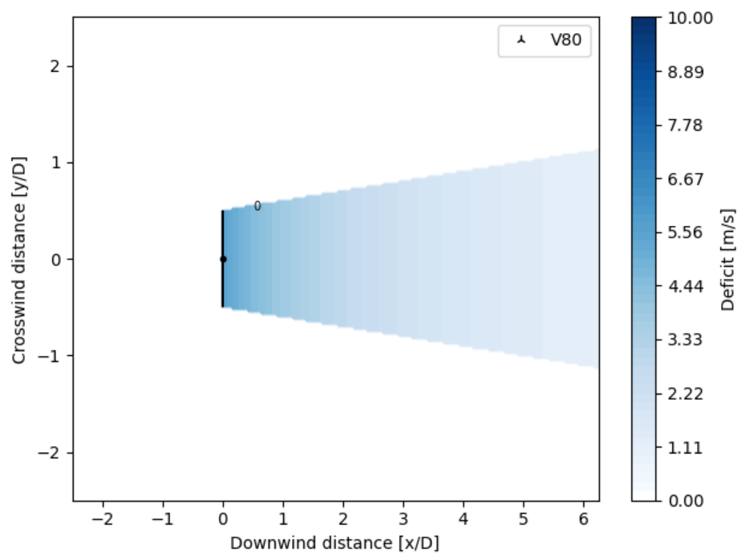


Figure 1: Illustration of the N.O. Jensen wake model [12].

$$r_w(x) = r_r + \alpha x \tag{4}$$

where r_r is the rotor radius, x is the downstream distance, and α is an empirical expansion coefficient. The wind speed in the wake is given by

$$u_w(x) = u_0 \left(1 - 2a\left(\frac{r_r}{r_r + \alpha x}\right)^2\right) \tag{5}$$

where u_0 is the free-stream wind speed and a is the axial induction factor. The thrust coefficient is defined as

$$C_T = 4a(1-a) \tag{6}$$

The velocity deficit is

$$\Delta u(x) = u_0 - u_w(x) \tag{7}$$

3.4.2 Gaussian Wake Model

More advanced wake models, such as the Gaussian wake model, describe the velocity deficit using a continuous distribution rather than a uniform profile [13]. The normalized velocity deficit is given by

$$\frac{\Delta u}{u_0} = \left(1 - \sqrt{1 - \frac{C_T}{8\left(\frac{\sigma}{D}\right)^2}}\right) \exp\left(-\frac{1}{2}\left(\frac{y}{\sigma}\right)^2\right) \tag{8}$$

where σ represents the wake width and increases with downstream distance. This model provides a more realistic description of the wake structure compared to simpler models.

3.4.3 Wake Deficit Superposition

In wind farms, turbines are often affected by multiple wakes. These must be combined using a superposition method. A commonly used approach is the squared sum method:

$$\Delta u_{\text{total}} = \sqrt{\sum_{i=1}^{N} (\Delta u_i)^2} \tag{9}$$

The effective wind speed is then

$$u = u_0 - \Delta u_{\text{total}} \tag{10}$$

This method provides a more realistic representation than simple linear summation and is widely used in wind farm modelling [11].

4 Methodology

4.1 Optimization problem

The optimization of wind farm layouts is a complex problem, as several factors influence the AEP. These include the wind resource, turbine characteristics, and wake effects between turbines. Based on the theoretical background presented in the previous section (section 3), the problem can be formulated as an optimization task where the goal is to determine the turbine positions that maximize the annual energy production (AEP).

4.1.1 Layout representation and decision variables

In this study, the wind farm layout is represented using a discretized grid of candidate positions within the site. The domain is divided into a grid, resulting in a specifeic number of potential turbine locations. This discretization defines the resolution of the optimization problem and directly influences its complexity and computational cost. The optimization problem is therefore formulated as a discrete selection problem, where the decision variables determine whether a candidate position is occupied by a turbine or not. Each candidate location i is associated with a binary variable b_i, where $b_i = 1$ if a turbine is placed at that position and $b_i = 0$ otherwise. The wind resource is defined through a site model, which includes a wind rose describing the distribution of wind directions, as well as wind speed distributions for each direction. These inputs are provided through the PyWake framework and are used to compute the AEP for each layout.

4.1.2 Constraints

The wind farm is restricted to a predefined boundary, obtained from the TOPFARM framework, ensuring that all selected turbine positions lie within the feasible site. In addition, a minimum spacing constraint is imposed between turbines to avoid excessive wake losses and mechanical interference. Furthermore, the total number of turbines is fixed, meaning that exactly k positions must be selected from the set of candidate locations.

4.1.3 Objective function

The objective of the optimization is to maximize the annual energy production (AEP) of the wind farm. The AEP is calculated based on the wind resource at the site, including both wind speed and direction distributions, as well as the interaction between turbines through wake effects. Wake effects reduce the effective wind speed experienced by downstream turbines and are therefore essential to include in the evaluation of each layout.

4.2 TOPFARM

To perform the layout optimization of the selected site, the TOPFARM framework was used. TOPFARM is an optimization tool developed at DTU, designed specifically for wind farm layout problems [3]. It allows the problem to be set up in terms of design variables, constraints, and an objective function [3]. The Main module is TopFarmProblem.

TopFarmProblems uses the constraint and varibales mentioned in the ealier section. In this study, the turbine positions are treated as the design variables. At the same time, constraints are applied to ensure that the turbines are placed within the defined boundary and with a minimum spacing between them. The annual energy production (AEP) is calculated using PyWake, which is integrated into TOPFARM [3] and used to evaluate how different layouts affect the overall energy production of the wind farm. PyWake is an open-source simulation tool used to model wind farm behavior by analyzing how turbine interactions affect airflow and overall energy production.

4.3 Models

In this study, several optimization models are applied to investigate wind farm layout optimization. The methods include heuristic, stochastic, and gradient-based approaches, each with different strengths in exploring and refining the design space. SmartStart is used to generate an initial layout based on a greedy heuristic. Random Search introduces stochastic perturbations to explore alternative configurations, while the 2-step greedy heuristic further improves the layout through iterative relocation of turbines. Finally, gradient-based optimization is applied to refine turbine positions in a continuous design space using gradient information.

Together, these models form a sequential optimization framework, where increasingly advanced optimization methods are applied to progressively refine the wind farm layout. SmartStart provides an initial feasible layout, Random Search introduces stochastic local exploration, the 2-step greedy heuristic performs systematic turbine relocation, and gradient-based optimization enables continuous local refinement using gradient information.

4.3.1 SmartStart

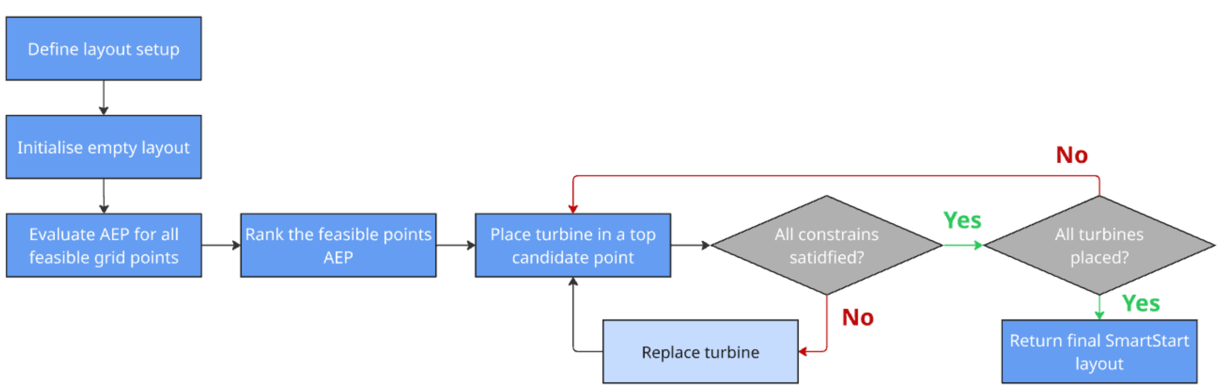


Figure 2: Flowchart of the SmartStart process.

To initialize the wind farm layout, the SmartStart algorithm is applied as the first optimization step. SmartStart is a greedy heuristic algorithm [5], which constructs the layout sequentially by iteratively placing turbines at positions that maximize the incremental Annual Energy Production (AEP). At each iteration, all feasible candidate positions are evaluated, and the most promising locations are identified before proceeding to the next turbine.

As illustrated in Figure Figure 2, the SmartStart algorithm follows an iterative procedure. First, a discretized mesh grid is defined to represent the set of candidate positions within the wind farm area. A minimum spacing constraint is imposed to ensure that turbines are placed at least a specified distance apart, typically defined relative to the rotor diameter.

The layout is initialized as empty, and at each iteration, all feasible candidate positions satisfying the spacing constraint are evaluated using the AEP model. The AEP is computed using the selected wake model and wind conditions through the `calc_aep()` function, accounting for wake interactions between turbines as well as wind direction and wind speed distributions.

After evaluating the feasible candidate positions, the points are ranked according to their resulting AEP values. Rather than always selecting the single best candidate, the algorithm randomly selects among the top-performing candidate positions to introduce stochasticity and improve exploration of the search space.

Based on the selected candidate point, a turbine is placed at the corresponding location. Once a turbine is placed, the selected position is removed from the candidate set together with all points violating the minimum spacing constraint. This progressively reduces the search space and ensures that all subsequent placements remain feasible. The layout is then updated, and the process is repeated until all turbines have been placed.

The level of stochasticity must be carefully balanced, as too high values lead to overly random layouts, while too low values result in a purely greedy and potentially locally optimal solution.

The resolution of the mesh grid plays an important role in the performance of the algorithm. A finer grid increases the number of candidate positions and may improve the solution quality, but at the cost of significantly higher computational effort [5].

Despite its simplicity, SmartStart offers high computational efficiency, allowing multiple optimization runs to be performed within reasonable time. This is particularly valuable for identifying general trends and ensuring robust comparisons between different scenarios. In contrast, more advanced optimization methods may achieve slightly better results in a single run but can introduce variability, where improvements may depend on favorable initial conditions rather than consistent performance.

While SmartStart provides a fast and effective method for generating layouts, it has inherent limitations. As a greedy heuristic, the algorithm optimizes turbine placement locally at each iteration without considering the global layout configuration. Furthermore, once a turbine is placed, its position cannot be adjusted in later iterations, which may prevent the algorithm from correcting suboptimal early decisions.

Therefore, SmartStart is primarily used to generate a strong initial layout, which is subsequently refined using more advanced optimization methods.

4.3.2 Random Search

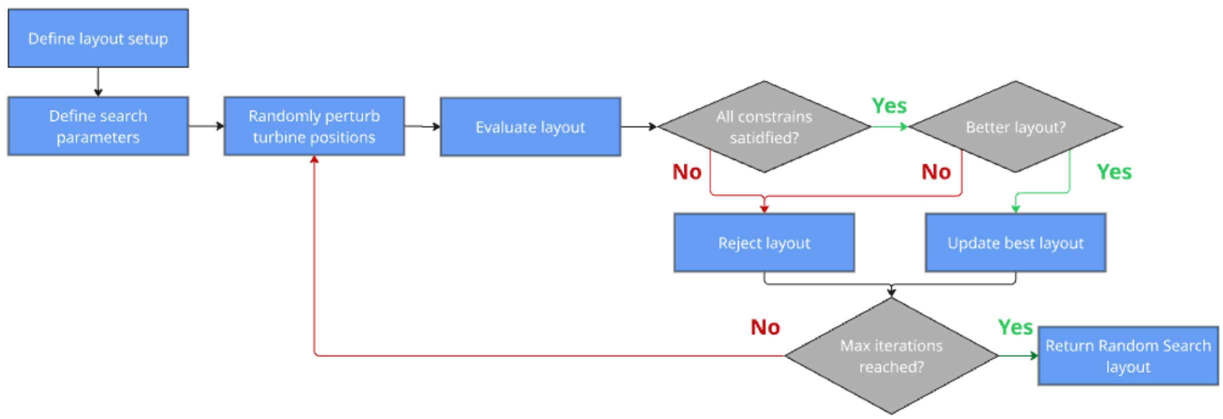


Figure 3: Flowchart of the Random Search process.

In addition to the heuristic SmartStart method, a stochastic optimization approach is applied in the form of Random Search. In this study, Random Search is implemented as a local search strategy that iteratively perturbs an existing wind farm layout.

As illustrated in Figure Figure 3, the algorithm is initialized using a predefined layout represented by the turbine coordinates. In addition, key search parameters are defined, including the maximum perturbation step size and the maximum number of iterations.

At each iteration, a new candidate layout is generated by randomly perturbing the turbine positions. The perturbation is bounded by a maximum step size defined relative to the rotor diameter, ensuring that turbine movements remain local and physically realistic.

After perturbation, the candidate layout is checked against the boundary and spacing constraints. Only feasible layouts are evaluated using the AEP model. The performance of the candidate layout is then compared to the best layout found so far. If the candidate yields a higher AEP, the best layout is updated. Otherwise, the candidate is rejected and the current best layout is retained.

In this way, the algorithm continuously tracks the best-performing solution throughout the optimization process. The procedure is repeated iteratively until a stopping criterion is reached, defined by the maximum number of iterations. The final output is the layout that achieved the highest AEP during the search.

Unlike grid-based methods such as SmartStart, Random Search operates in a continuous design space, allowing turbine positions to move freely within the defined constraints. This enables a more flexible exploration of the solution space. However, as a stochastic method, Random Search does not guarantee convergence to a global optimum and may require a large number of iterations to achieve significant improvements.

4.3.3 2-step greedy heuristic

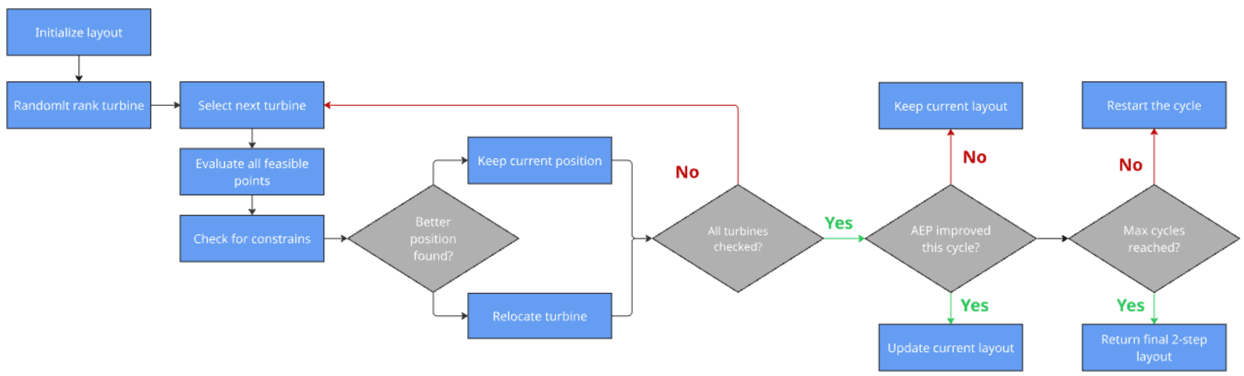


Figure 4: Flowchart of the 2-step greedy heuristic (Stage 2). Each turbine is iteratively relocated by evaluating feasible positions, and improvements are accepted based on the total AEP.

As mentioned in the previous sections, SmartStart and Random Search do not guarantee a globally optimal solution. Additional optimization methods are therefore introduced. The 2-step greedy heuristic is used to further improve an existing layout obtained from SmartStart or Random Search.

As illustrated in Figure Figure 4, the algorithm starts from an initial layout and iteratively improves the position of each turbine. At the beginning of each cycle, the turbines are processed in a randomized order to reduce deterministic search behavior and improve exploration of the solution space.

One turbine is selected at a time and temporarily removed from the layout. All feasible candidate grid points satisfying the spacing constraint are then evaluated as potential replacement positions. For each candidate position, the total AEP of the wind farm is calculated. This ensures that the impact of relocating a turbine is evaluated based on the full interaction between all turbines, including wake effects.

If a candidate position yields a higher AEP than the current turbine position, the turbine is relocated. Otherwise, the current position is retained. This process is repeated for all turbines in the wind farm, completing one full optimization cycle.

After each cycle, the algorithm checks whether the total AEP has improved compared to the previous cycle. If an improvement is observed, another cycle is initiated. If no improvement is found, or if a predefined maximum number of cycles is reached, the algorithm terminates.

Throughout the optimization process, wake interactions are continuously updated whenever turbine positions change. This ensures that each relocation step reflects the global effect on the wind farm performance.

Compared to SmartStart and Random Search, the 2-step greedy heuristic refines an existing layout through systematic turbine relocation. This allows the algorithm to correct suboptimal placements introduced during the initial optimization stage and results in a more globally improved configuration.

As a result, the method improves convergence towards higher AEP values while maintaining a relatively low computational cost compared to more advanced optimization techniques [5].

4.3.4 Gradient-based optimization

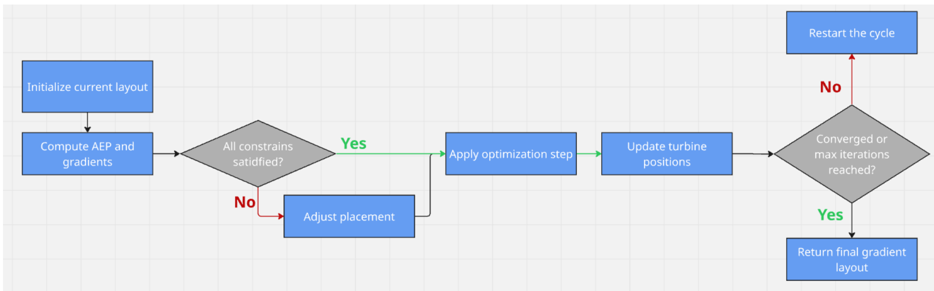


Figure 5: Flowchart of the gradient-based optimization process. Turbine positions are iteratively updated using gradient information while satisfying boundary and spacing constraints.

Another approach for maximizing the Annual Energy Production (AEP) is the use of gradient-based optimization methods applied after an initial layout has been generated. In this context, gradient-based optimization serves as a refinement step, where an existing turbine configuration is further improved by adjusting turbine positions in a continuous design space. Unlike discrete optimization methods that select positions from a predefined set of candidate locations, gradient-based methods operate directly on the continuous (x, y) coordinates of each turbine [14].

As illustrated in Figure Figure 5, the optimization is initialized from a given layout, which serves as the starting point for the continuous optimization. The turbine coordinates are treated as design variables and are iteratively updated to maximize the AEP.

At each iteration, the objective function, defined as the total AEP, is evaluated together with its gradients with respect to the turbine positions. The gradients indicate the direction of the steepest increase in AEP and are computed using automatic differentiation within the PyWake framework.

Based on the computed gradients, the optimizer applies an optimization step that updates the turbine coordinates in the direction of increasing AEP. In this study, the optimization is implemented using the Sequential Least Squares Programming (SLSQP) algorithm available in TOPFARM.

Boundary and spacing constraints are incorporated directly into the optimization problem, ensuring that the proposed turbine movements remain feasible throughout the optimization process. If constraint violations occur, the optimizer applies constraint corrections before continuing the optimization.

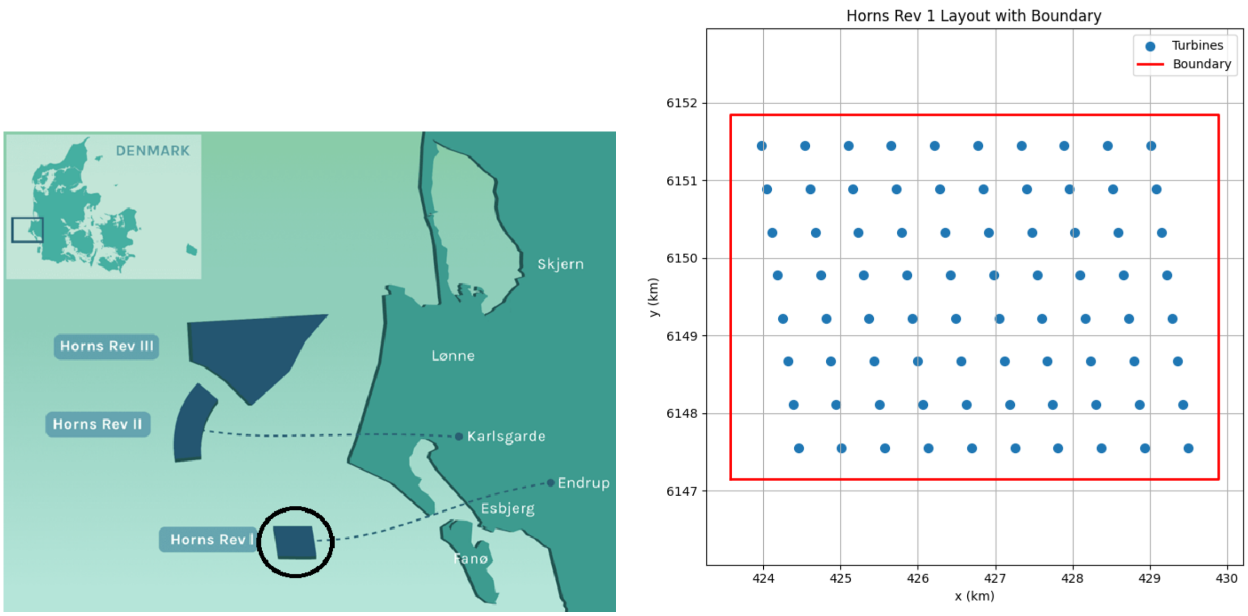
The iterative process continues until a convergence criterion is reached, such as a negligible improvement in AEP or a predefined maximum number of iterations. The final output is the optimized turbine layout with the highest feasible AEP found during the optimization process.

Compared to heuristic methods, gradient-based optimization enables precise local refinement by directly exploiting gradient information. However, the method is sensitive to the initial layout and may converge to local optima.

5 Model setup

5.1 Case Setup

The case study is based on the Horns Rev 1 offshore wind farm, which is commonly used as a reference site for wind farm layout optimization studies, due to the simplicity of the layout. The site provides realistic wind conditions and turbine data, making it suitable for benchmarking optimization approaches.



(a) Map of Horns Rev with location of Horns Rev 1 [15]

(b) Layout of Horns Rev 1 with boundary

Figure 6: Overview of the Horns Rev 1 wind farm location and layout

The wind rose shown in Figure 7 illustrates the distribution of wind directions at the site. It can be observed that certain wind directions occur more frequently, which has a significant impact on wake interactions between turbines. In this study, the wind rose is defined at the site level and applied uniformly across the entire wind farm. This means that all turbines are assumed to experience the same wind direction distribution, rather than accounting for local variations within the farm. As a result, the layout optimization must account for these dominant wind directions in order to maximize the overall AEP. The wind speed distribution is based on site-specific data provided by PyWake and is applied uniformly across all turbine positions. As a result, the layout optimization must account for these dominant wind directions and the overall wind resource characteristics in order to maximize the AEP. The wind farm consists of $N_{WT} = 80$ wind turbines. The layout domain is discretized into a grid of candidate

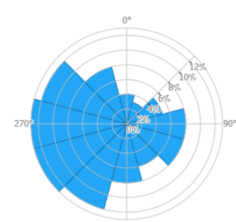


Figure 7: Wind rose for Horns Rev 1 showing the directional distribution of wind frequencies [16]

positions in order to reduce the search space. In this study, a standard grid of 20×20 points is used, resulting in a total of 400 potential turbine locations. This discretization defines the resolution of the layout optimization problem and directly influences both computational cost and solution quality. A reference wind turbine is used to represent the turbine characteristics. The turbine corresponds to the V80 model, with a rotor diameter of $D = 80$ m and a rated power of approximately 2 MW. These parameters define both the power production and the wake interactions between turbines. The power and thrust characteristics of the reference turbine are illustrated in Figure 8, showing how the power output and thrust coefficient vary with wind speed.

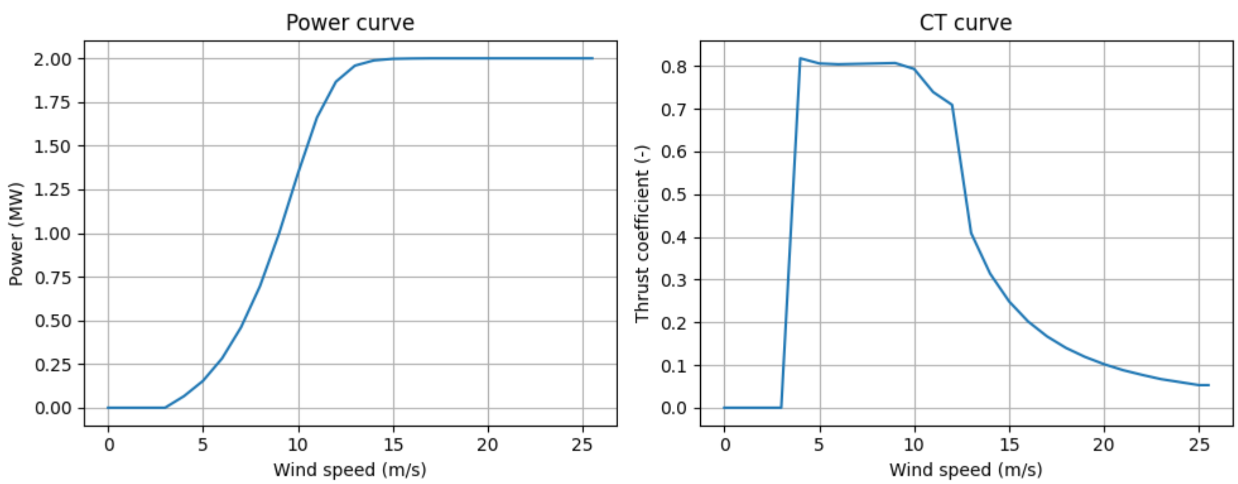


Figure 8: Power curve (left) and thrust coefficient (CT) curve (right) for the reference wind turbine as a function of wind speed.

Power curve (left) and thrust coefficient (CT) curve (right) of the V80 reference wind turbine as functions of wind speed. The power curve shows the increase in power output up to the rated region, while the CT curve illustrates how the turbine loading and wake-inducing effects vary with wind speed. A minimum spacing constraint is applied to ensure realistic turbine placement and to avoid excessive wake losses. The spacing is defined relative to the rotor diameter and is set to $4D$, meaning that the distance between any two turbines must be at least four times the rotor diameter. The wind resource at the site is described using directional wind distributions, where both wind speed and wind direction are considered in the AEP calculation. The evaluation of energy production is performed using a wake model, which accounts for the interaction between turbines and the resulting wake losses. The discretized grid, turbine specifications, and spacing constraints together define the design space of the optimization problem.

5.2 Iterations and bins

Before explaining the different modules and programs, it is important to describe some of the key factors and computational aspects used in this thesis. When using PyWake together with site specifications from TOPFARM, the wind resource is already discretized into bins of wind speed and wind direction. Typically, wind speeds are divided into 1 m/s bins, while wind directions are divided into 10° intervals, resulting in 36 directional bins. This leads to approximately 8,000 AEP evaluations, as the AEP is calculated using the summation $\sum_{d=1}^{N_\theta} \sum_{u=1}^{N_u} \sum_{n=1}^{N_{WT}}$ as shown in

Equation 3. When combined with the iterative optimization process, using a 20×20 grid, 80 turbines, and 10 cycles, this results in

$$400 \cdot 80 \cdot 10 = 320{,}000 \tag{11}$$

layout evaluations. Each of these evaluations requires multiple AEP calculations across all bins, making the computational cost extremely high, but also yielding a highly accurate estimation of the wind farm performance. To address this, simplifications are introduced in this study. While the full bin representation provides a realistic estimate of AEP, it is computationally expensive. Therefore, inspired by [4], the number of bins is reduced by using a single mean wind speed for all wind directions, while maintaining a discretization of wind directions into 5° bins. This significantly reduces the computational cost, as also observed in [4], where runtime reductions of up to 60% are reported. However, this simplification comes at the cost of reduced accuracy. The resulting AEP values should therefore not be interpreted as the true annual energy production of the wind farm, but rather as a surrogate AEP used to compare the relative performance of different layouts. By using a consistent approximation across all optimization runs, the relative improvements remain meaningful, while the computational effort is significantly reduced.

This simplification neglects the variability in wind speed as well as the nonlinear relationship between wind speed and power production. However, this trade-off is accepted within the scope of this project, where the primary objective is to compare different optimization methods. In this context, computational efficiency is essential, and reducing the number of bins significantly decreases the runtime of the AEP evaluations.

5.3 Wake Model

The interaction between wind turbines is modeled using the N.O. Jensen wake model (subsubsection 3.4.1). The model is selected due to its computational efficiency, making it well suited for optimization problems requiring a large number of AEP evaluations. Wake interactions are combined using the squared sum superposition method, ensuring a consistent representation of multiple wake effects throughout the optimization. The choice of the Jensen model reflects the focus of this study, which is on comparing optimization strategies rather than achieving the highest possible wake model accuracy. Although more advanced models, such as Gaussian wake models, can provide improved physical representation, they are also more computationally demanding. To evaluate the impact of the wake model choice, a comparison with more advanced wake models is included in the sensitivity analysis. All wake calculations are performed using the PyWake framework. Eventuelt indsæt plot med sammenligning af NO jensen og Gaussian i forhold til baseline med SmartStart - 2-step - gradient for at vise forskellen

5.4 AEP evaluation

The performance of each wind farm layout is evaluated based on the Annual Energy Production (AEP). The AEP is calculated using the `PyWakeAEPCostModelComponent` implemented in TOPFARM. This component evaluates the energy production of the wind farm based on the turbine positions, the site-specific wind resource, and the selected wake model. For each layout, the power production is computed across the mean wind speed and wind direction bins, and the total AEP is obtained by summing the contributions from all turbines and wind conditions. Using a consistent AEP evaluation method ensures that differences in performance between pipelines are solely due to the optimization strategies and not variations in the evaluation setup.

5.5 Model parameters

In this study, 1 initial layout algorithm and 3 optimization approaches are considered. These methods are used either individually or in combination to improve the wind farm layout. For a detailed description the different models see section 4. Each model has different Constraints or parameter to define the work done by each model.

5.5.1 Constraints

Type	**Name / Symbol**	**Description**
Design variable	x	Turbine x-coordinate
Design variable	y	Turbine y-coordinate
Constraint	Boundary constraint	Turbines must remain within the defined wind farm boundary
Constraint	Spacing constraint	Minimum distance of $4D$ between turbines
Constraint	Number of turbines	The layout must contain $N_{WT} = 80$ turbines
Objective	AEP	Maximize Annual Energy Production

Table 1: Common design variables and constraints used across all optimization models.

In Table 1 we see the common design variables and constrain to define the initial layout. These define the feasible design space within which the layout is optimized to maximize the Annual Energy Production (AEP).

SmartStart

Parameter	**Description**
x_points	Grid points in x-direction
y_points	Grid points in y-direction
random_pct	Random selection among top candidates
seed	Random seed

Random Search

Parameter	**Description**
max_step	Maximum turbine displacement
max_iter	Maximum number of iterations
max_time	Maximum runtime
seed	Random seed

2-step greedy heuristic

Parameter	**Description**
max_cycles	Maximum relocation cycles
x_points	Candidate grid points in x-direction
y_points	Candidate grid points in y-direction

Gradient-based optimization

Parameter	**Description**
optimizer	SLSQP optimizer
grad_method	Automatic differentiation
max_iter	Maximum number of iterations
tolerance	Convergence criterion

Table 2: Key model-specific parameters used in the four optimization models.

The model-specific parameters used in each optimization method are summarized in Table 2. These parameters define how each algorithm explores and refines the design space, influencing both the convergence behaviour and the resulting AEP. Some parameters have a significant influence on the behaviour of the optimization algorithms. In SmartStart and the 2-step method, the grid resolution determines the number of candidate positions and therefore affects both computational cost and solution accuracy. The parameter `random_pct` in SmartStart introduces stochasticity, allowing the algorithm to explore multiple near-optimal placements. For Random Search, the parameter `max_step` controls the magnitude of turbine perturbations, while `max_iter` and `max_time` determine the extent of the search. In the 2-step method, `max_cycles` defines the number of refinement iterations and thus influences convergence. In gradient-based optimization, the choice of optimizer and convergence tolerance determines how precisely the turbine positions are refined and when the algorithm terminates.

5.5.2 Simulation inputs

In addition to the model-specific parameters, a common set of simulation inputs is used across all optimization runs. These inputs define the wind conditions, grid resolution, spacing requirement, boundary padding, and wake model used in the AEP evaluation.

Type	Name / Symbol	Description
Wind	*ws_mean*	Mean wind speed [m/s]
Wind	*wd_bins*	Wind direction bins [deg]
Grid	*x_points*, *y_points*	Grid resolution
Spacing	$spacing_D$	Minimum spacing in rotor diameters
Boundary	boundary_pad	Padding around site boundary [m]
Model	wake_model	Selected wake model (NOJ / Gaussian)

Table 3: Wind and simulation parameters used in the optimization.

5.6 Pipelines

The scope of this project is to compare different optimization pipelines for wind farm layout design. The pipelines are based on two different initial layout approaches: SmartStart and Random Search. These initial layouts are either evaluated directly or further refined using the 2-step greedy heuristic and/or gradient-based optimization. The SmartStart method utilizes wind speed and wind direction data from the Horns Rev 1 site to generate an initial turbine layout. This layout serves as the baseline for all subsequent optimization steps, ensuring a consistent starting point for comparison. Following the initial layout generation, additional optimization methods are applied to further improve the Annual Energy Production (AEP). These include a discrete 2-step optimization approach and a continuous gradient-based refinement. The pipelines considered in this study are defined as follows:

Table 4: Optimization pipelines

No.	Abbreviation	Pipeline
1	SS	SmartStart
2	SS–2S	SmartStart → 2-step
3	SS–GB	SmartStart → Gradient-based
4	SS–2S–GB	SmartStart → 2-step → Gradient-based
5	RS	Random Search
6	RS–2S	Random Search → 2-step
7	RS–GB	Random Search → Gradient-based
8	RS–2S–GB	Random Search → 2-step → Gradient-based

6 Results and analysis

This chapter presents and analyzes the results obtained from the different wind farm layout optimization methods. First, the Horns Rev 1 reference layout is introduced as the common baseline for comparison. Subsequently, the performance of the different optimization pipelines is evaluated using both the NOJ and Gaussian wake models. Finally, a sensitivity analysis is conducted to investigate the influence of important optimization parameters and model assumptions.

6.1 Sensitivity Analysis

This section investigates the sensitivity of the optimization results to different parameters and model assumptions. All runs has been made with the same parameters:

Table 5: Common baseline parameters used across the optimization studies. Individual sensitivity analyses modify selected parameters as specified in their respective sections.

Parameter	Value
Wind farm site	Horns Rev 1
Number of turbines, n_{wt}	80
Wake model	NOJ
Minimum turbine spacing	$4D$
Candidate grid resolution	20×20
Boundary padding	400 m
Mean wind speed	9.6 m/s
Wind direction discretization	5° bins

The parameters listed in Table 5 represent the baseline setup used throughout the study. Selected parameters are varied within the individual sensitivity analyses depending on the investigated effect.

6.1.1 SmartStart `random_pct`

This sensitivity study investigates how the SmartStart `random_pct` parameter influences the optimization performance and convergence behavior. The `random_pct` parameter controls the degree of stochasticity during turbine placement by selecting the next turbine position randomly among the top-performing candidate locations instead of always selecting the single best candidate. A value of 0% therefore corresponds to a fully deterministic SmartStart optimization, while larger values increase the level of random exploration.

The SmartStart algorithm was executed using `random_pct` values of 0%, 3%, 5%, 10%, 20%, and 50% while keeping all other optimization parameters constant. Each case was repeated using three different random seeds, and the presented results show the mean behavior across the three runs.

Table 6: Sensitivity analysis of the SmartStart `random_pct` parameter

random_pct	Mean AEP [GWh]	Best AEP [GWh]	Std. Dev.	Seed count
0%	72.395	72.448	0.053	3
3%	72.272	72.305	0.043	3
5%	72.106	72.111	0.009	3
10%	72.098	72.122	0.028	3
20%	71.990	72.102	0.098	3
50%	71.553	71.683	0.126	3

Table 6 and Fig. 9 summarize the results.

The results show that the deterministic SmartStart configuration with `random_pct=0%` achieves the highest mean and best AEP for the present optimization setup. Increasing the level of stochasticity gradually reduces the optimization performance, with the largest degradation observed for the 50% case. In addition, the standard deviation between the random seeds generally increases with larger `random_pct` values, indicating reduced optimization robustness and larger variation between generated layouts.

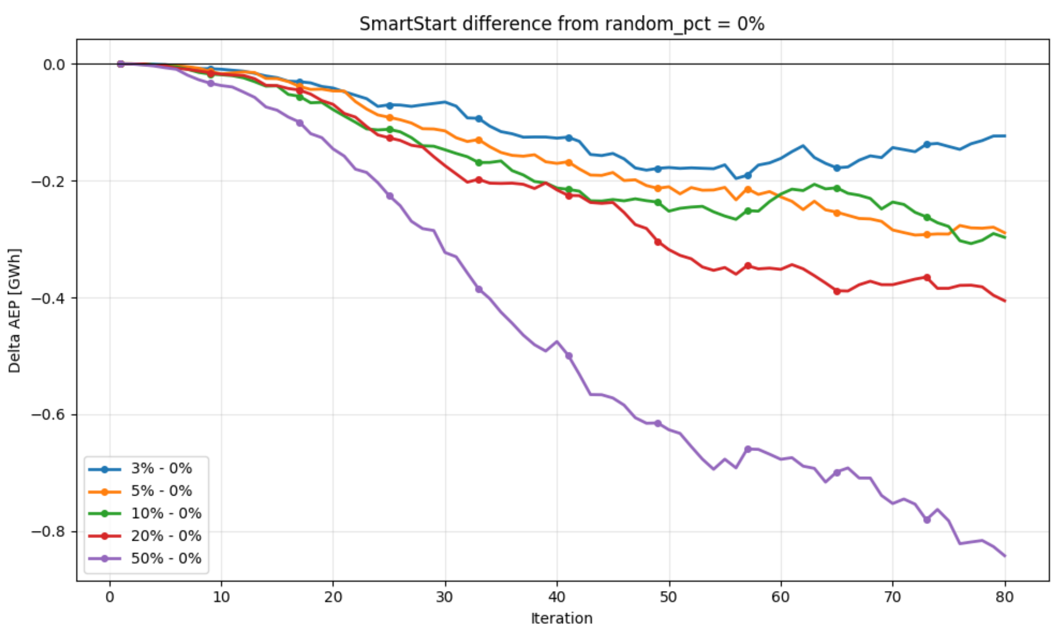


Figure 9: Difference in AEP relative to the deterministic SmartStart case (`random_pct=0%`) during the sequential turbine placement process.

Fig. 9 further illustrates how the influence of the `random_pct` parameter accumulates throughout the sequential turbine placement process. During the first turbine placements, all cases remain relatively close to the deterministic reference. However, as additional turbines are placed, the stochastic cases increasingly diverge from the deterministic solution. This indicates that small differences introduced during the early placement stages propagate and influence the remaining layout construction process. The 3%, 5%, and 10% cases remain relatively close to the deterministic solution throughout the optimization, suggesting that limited stochasticity only slightly alters the placement strategy. In contrast, the 20% and especially the 50% cases show a substantially larger reduction in AEP as the optimization progresses. The 50% configuration exhibits a nearly continuous degradation relative to the deterministic reference, ultimately resulting in a significantly lower final AEP. These results indicate that excessive stochasticity weakens the greedy nature of the SmartStart algorithm. Since turbine placement becomes increasingly random, the algorithm is less capable of consistently selecting the most favorable candidate positions during the sequential layout construction process. For the relatively simple and homogeneous offshore wind farm considered in this study, the deterministic SmartStart approach therefore appears to be the most effective strategy. The optimization problem is dominated primarily by wake interactions and spacing effects, making the greedy placement strategy highly effective. Nevertheless, the `random_pct` parameter may become more valuable for more complex optimization problems. In wind farms with irregular boundaries, terrain-induced flow variations, varying roughness conditions, or more constrained feasible regions, a fully deterministic greedy placement may become trapped in suboptimal placement patterns. Introducing limited stochasticity may therefore improve exploration of the design space and increase layout diversity across multiple optimization runs. Another important purpose of the `random_pct` parameter is to increase layout diversity when the optimization is repeated using different random seeds. Since the turbine placement is randomly selected within the percentage range defined by `random_pct`, each optimization run may result in a slightly different layout. This increases the probability that a larger portion of the solution space is explored, thereby improving the likelihood that improved layout configurations are identified across multiple optimization runs.

6.1.2 RandomSearch Convergence Behavior

This sensitivity study investigates the convergence behavior of the RandomSearch algorithm and evaluates how the optimization progresses over time. The RandomSearch optimization was executed for 10000 iterations using the NOJ wake model while keeping all other optimization parameters constant.

To reduce the influence of stochastic behavior, the optimization was repeated using six different random seeds. Figure 10 presents the mean AEP convergence across all runs, while the shaded region illustrates the variation between the different seeds.

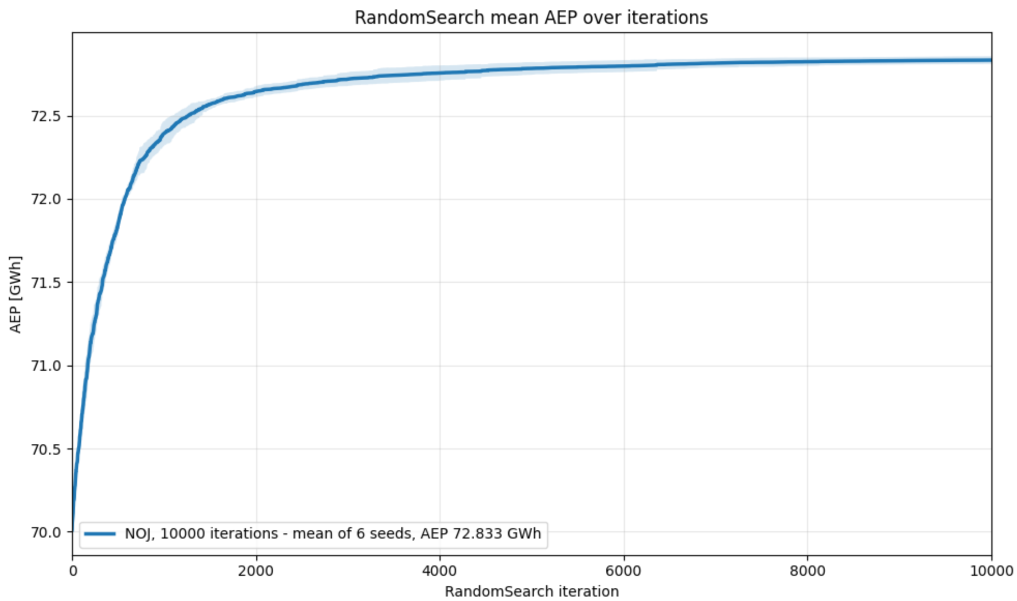


Figure 10: Mean AEP convergence of the RandomSearch algorithm over 10000 iterations using the NOJ wake model. The shaded region indicates the variation across six random seeds.

The convergence curve shows that the RandomSearch algorithm achieves rapid improvements during the initial stages of the optimization. A large portion of the total AEP improvement is obtained within the first 1000–2000 iterations, after which the convergence rate gradually decreases.

Beyond approximately 2000 iterations, only minor AEP improvements are observed despite the continued increase in computational effort. The convergence behavior therefore indicates diminishing returns for larger iteration counts, suggesting that the optimization approaches a near-stationary solution after the early search stages.

Although the final iterations still provide marginal improvements, the results indicate that an iteration limit around 2000 iterations provides a reasonable balance between optimization quality and computational cost for the present study. Consequently, this iteration limit was selected for the remaining RandomSearch analyses in this thesis.

6.1.3 RandomSearch Step Size

This sensitivity study investigates how the maximum RandomSearch step size influences the convergence behavior, final layout quality, and computational runtime. The RandomSearch algorithm was executed using maximum step sizes of 1D, 5D, 10D, 20D, 30D, and 50D while keeping all other optimization parameters constant. The step size defines the maximum distance a turbine is allowed to move during each stochastic perturbation and is specified relative to the turbine rotor diameter, *D*.

Each step-size case was repeated using three different random seeds, and the presented convergence and runtime plots show the mean behavior across the three runs. This reduces the influence of stochastic variation and provides a more representative comparison between the tested step sizes.

Convergence Behavior

Fig. 11 shows the AEP convergence as a function of elapsed runtime for the different RandomSearch step sizes.

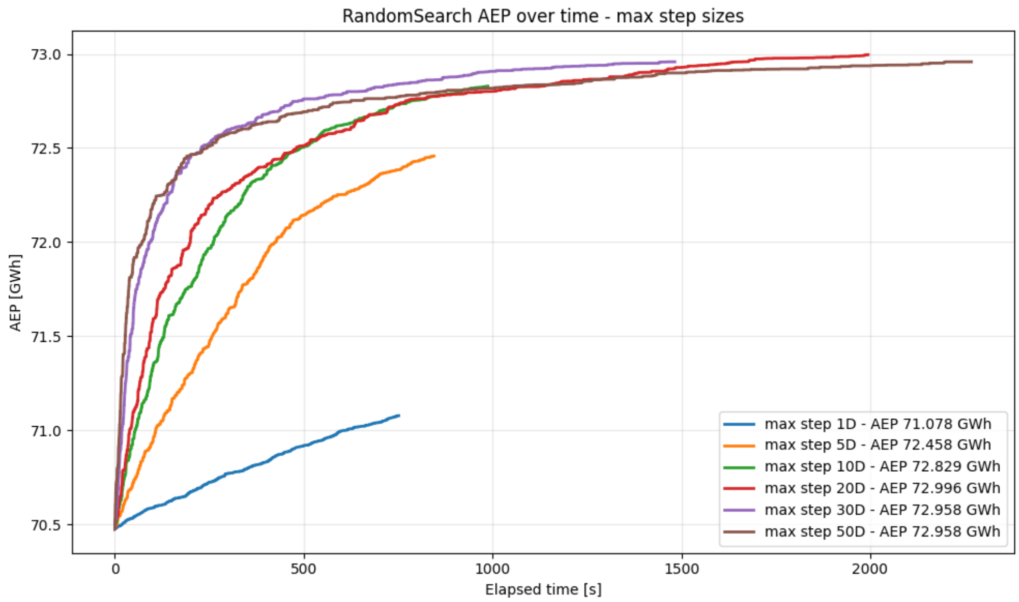


Figure 11: AEP progression as a function of elapsed runtime for different RandomSearch step sizes using the NOJ wake model.

The convergence curves show that the selected step size has a strong influence on the optimization behavior. Very small step sizes, such as 1D and 5D, mainly perform local exploration around the current layout. This results in relatively slow convergence and limits the ability of the algorithm to significantly restructure the wind farm layout. In particular, the 1D case remains trapped close to the initial configuration and converges towards a substantially lower final AEP compared to the larger step sizes.

As the step size increases, the algorithm becomes capable of performing larger layout modifications, enabling a broader exploration of the design space. The 10D, 20D, 30D, and 50D cases all show significantly faster initial improvements in AEP and converge towards considerably higher final layouts. However, the results also indicate diminishing returns for very large step sizes. While the 30D and 50D cases continue to improve slightly at later stages of the optimization, the improvement compared to the 20D case remains relatively small despite the increased runtime.

The convergence behavior also suggests that different step sizes are advantageous at different stages of the optimization process. Large step sizes are particularly effective during the early optimization stages, where large structural layout changes are beneficial for escaping the initial configuration and exploring the global design space. In contrast, smaller step sizes are better suited for the later optimization stages, where the layout is already relatively optimized and the search shifts towards local refinement.

Based on these observations, it may therefore be beneficial to combine multiple step sizes within the same optimization run. Such a staged optimization strategy could begin with large step sizes to encourage global exploration and subsequently reduce the step size to perform finer local adjustments near convergence. To investigate this behavior, an additional staged step-size schedule was introduced, consisting of 1200 iterations using 20D, followed by 400 iterations with 5D and finally 400 iterations with 1D.

Final AEP Performance

Fig. 12 shows the final AEP obtained for each step-size configuration, including the staged step-size schedule.

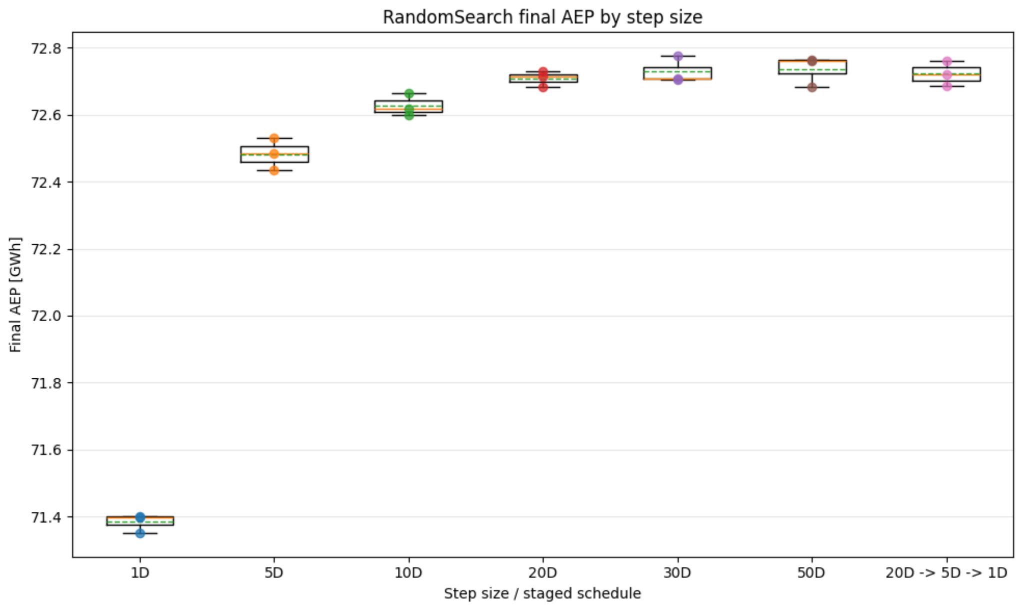


Figure 12: Final AEP obtained for different RandomSearch step sizes and the staged step-size schedule. Each case is repeated using three random seeds.

The results show that increasing the step size from 1D to 20D significantly improves the final AEP. This confirms that larger perturbations are necessary to effectively explore the design space and avoid premature convergence near the initial layout. Beyond 20D, the AEP improvements become smaller, indicating diminishing returns for increasingly large perturbations.

The staged step-size schedule achieves a final AEP comparable to the best fixed step-size cases. This suggests that combining large initial exploration with smaller local refinement can provide an effective balance between global search and local optimization. Although the staged schedule does not clearly outperform the best fixed step-size configuration, it demonstrates that adaptive step-size strategies may improve the robustness and efficiency of the optimization process.

Runtime Scaling

Fig. 13 shows the computational runtime for the different RandomSearch step sizes.

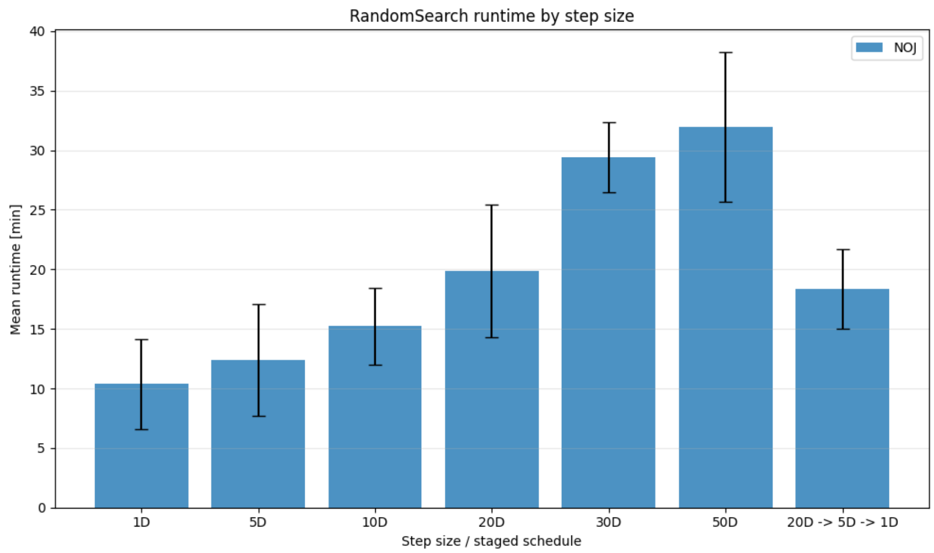


Figure 13: Mean runtime for each RandomSearch step-size configuration. Error bars indicate variation between the three random seeds.

The runtime generally increases with the selected step size. Larger perturbations allow the algorithm to explore more distinct layouts, which increases the number of accepted improvements and prolongs the optimization process. In contrast, smaller step sizes produce more localized changes and therefore shorter runtimes, but at the expense of reduced optimization performance.

Interestingly, the staged step-size schedule achieves a runtime closer to the 20D case while still obtaining a final AEP comparable to the larger step-size configurations. This indicates that staged optimization strategies may provide a reasonable compromise between computational cost and layout quality.

Overall, the results demonstrate that the RandomSearch step size strongly influences the balance between local refinement and global exploration. Very small step sizes lead to premature local convergence, while excessively large step sizes increase runtime without substantial additional AEP improvements. For the present optimization setup, step sizes in the range of 20D–30D appear to provide the most favorable balance between convergence behavior, final AEP, and computational efficiency.

6.1.4 Wind Direction Bin Study

This sensitivity study investigates how the discretization of wind direction influences the computational runtime of the SmartStart and two-step optimization pipeline. The study is based on four different wind direction resolutions using a fixed mean wind speed of 9.6 m/s. The tested configurations use wind direction bins of 10°, 5°, 2°, and 1°, corresponding to 36, 72, 180, and 360 wind direction sectors, respectively. It is important to note that the purpose of this study is primarily to investigate computational runtime and convergence behavior rather than direct AEP comparison. Since the wind direction discretization changes the representation of the wind climate used in the AEP calculation, the resulting AEP values are not directly comparable across all cases.

Table 7: Sensitivity analysis of wind direction discretization

Case	Size of direction bins	Number of sectors
360wd_mean_ws	1°	360
180wd_mean_ws	2°	180
72wd_mean_ws	5°	72
36wd_mean_ws	10°	36

Fig. 14 shows the mean runtime over iterations for the four wind direction discretizations. Each case was repeated using three different random seeds, and the plotted curves show the mean runtime behavior across the runs.

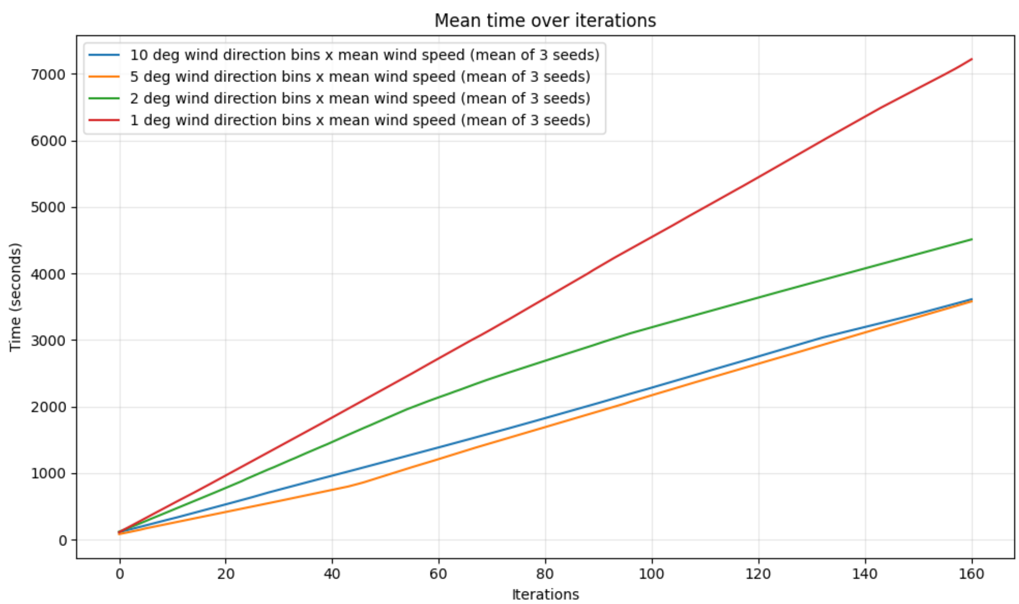


Figure 14: Mean runtime over iterations for different wind direction discretizations using a fixed mean wind speed.

The results show a clear relationship between the number of wind direction sectors and the computational runtime. Increasing the wind direction resolution significantly increases the computational cost of the optimization. The 1° case requires the highest runtime, while the 10° case remains the most computationally efficient. This behavior is expected, since increasing the number of wind direction sectors increases the number of wake model evaluations required during each AEP calculation. As a result, each optimization iteration becomes more computationally expensive. The runtime increase is approximately linear with the number of wind direction sectors, particularly for the finer discretizations. The results also show that the runtime difference between the 5° and 10° cases remains relatively moderate compared to the substantial increase observed for the 2° and 1° cases. This indicates that very fine wind direction discretizations rapidly increase the computational cost while providing diminishing practical benefits for optimization efficiency. Overall, the study demonstrates the trade-off between computational efficiency and wind climate resolution. Finer wind direction discretizations provide a more detailed representation of the wind resource and wake interactions, but at the cost of substantially increased runtime. For the present study, the use of 5° wind direction bins combined with a mean wind speed representation provides a reasonable compromise between computational efficiency and representation of the wind climate. Using excessively large wind direction bins may lead to inaccurate AEP estimations, since the directional resolution becomes too coarse to properly capture variations in wake interactions and wind inflow conditions between neighboring wind sectors.

6.1.5 Wake Model Sensitivity

Table 8: Comparison of wake model sensitivity

Wake model	Pipeline	AEP [GWh]	Runtime [min]	Relative difference [%]
NOJ				
Gaussian				

Discuss how wake model assumptions influence the optimized layouts and AEP predictions.

6.1.6 Seed Sensitivity

Table 9: Sensitivity analysis of random seed initialization

Method	Mean AEP	Best AEP	Worst AEP	Std. Dev.	Seed count
SS					
RS					
SS–GB					
RS–GB					

Discuss the robustness and reproducibility of the optimization methods.

6.2 Summary of Results

Summarize the main findings from the optimization comparison and sensitivity analysis. Highlight the best-performing optimization pipeline, the most important parameter sensitivities, and the observed trade-offs between optimization quality and runtime.

7 Discussion

7.1 Initialization Strategy

When comparing SmartStart and Random Search as initialization strategies, an important challenge arises regarding the fairness of the comparison. Is it even possible to compare the two methods? The two methods are fundamentally different in their optimization approach. SmartStart is a constructive heuristic that generates a layout directly from the site definition and optimization constraints, while Random Search requires an existing predefined layout as its starting point, such as the Horns Rev 1 reference layout. This creates a fundamental difference between the two initialization strategies. SmartStart constructs a layout sequentially from an initially empty wind farm, whereas Random Search improves an already existing layout through stochastic perturbations. As a result, a direct one-to-one comparison between the methods becomes difficult. To establish a more comparable framework, the Random Search algorithm was tuned to allow sufficiently large turbine movements such that the layout could evolve significantly away from the initial Horns Rev 1 configuration. This enabled the algorithm to move turbines towards the wind farm boundaries and gradually form layouts similar to those generated by SmartStart. But this trade-off to make them comparable comes with a prize. As observed in the results, Random Search required significantly more iterations and computational runtime to obtain layouts comparable to SmartStart. However, this also highlights a key difference between the methods. SmartStart generates a single greedy layout in one optimization sequence, while Random Search evaluates thousands of candidate layouts throughout the optimization process. In this sense, Random Search gains an "head start" by repeatedly sampling the design space, which also explains why the resulting layouts often provided stronger starting points for the later optimization stages. This difference is further reflected in the performance of the 2-step greedy refinement. The layouts generated by SmartStart required approximately 9 cycles on average before convergence, whereas layouts initialized using Random Search converged after approximately 7 cycles. This indicates that the additional exploration performed by Random Search produced layouts that were already closer to a locally optimized configuration before entering the refinement stage. Another important aspect is the complexity of the optimization problem itself. In the present study, the optimization is performed on a relatively simple and uniform offshore wind farm layout with similar wind and surface conditions across the entire site. Under these conditions, the deterministic and computationally efficient nature of SmartStart makes the algorithm highly effective. However, for more complex wind farm layouts, such as irregular offshore boundaries or onshore wind farms with varying terrain roughness and obstacles, a purely greedy initialization may become less robust. As discussed in subsubsection 6.1.1, the stochastic component introduced through the random percentage parameter allows SmartStart to generate more diverse layouts across multiple runs, thereby improving exploration of the design space. In comparison, Random Search continuously perturbs turbine positions throughout the optimization and therefore explores a significantly larger portion of the feasible design space. While this comes at the cost of substantially higher computational runtime, it also increases the probability of identifying improved layouts in optimization problems with higher geometric and aerodynamic complexity. Overall, the results suggest that SmartStart provides a highly computationally efficient initialization strategy for relatively uniform offshore wind farm layouts, whereas Random Search offers greater exploratory capability at the expense of computational cost.

7.2 Hybrid Optimization Approaches

Discuss the benefits and limitations of combining discrete and continuous optimization approaches. This includes how SmartStart and Random Search generate initial layouts, how the 2-step heuristic performs large structural improvements, and how gradient-based optimization mainly provides local refinement.

This study is to combine different optimazation methods into larger hybrid pipelines. The motivation for this approach come from ready the article about the two step greedy algorithm. In the litterature review (See section 2) the different methods are well decsriped but not many studies uses more than one optimazation method. In **??** the different pipelines illustrates that with this approach of a hybrid pipelines is somewhat sufficient, with a increase in the objection function AEP. This comes with a tradeoff in computional effiency

7.3 Runtime versus Performance

Discuss the trade-off between computational cost and optimization quality. This includes the influence of wind direction discretization, candidate grid resolution, Random Search iterations, and simplified wind climate representations. The section can also discuss diminishing returns, where large increases in runtime only provide limited improvements in AEP or layout quality.

7.4 Limitations

The present study includes several simplifications and limitations that should be considered when interpreting the results.

- Simplified wake models
- Single-site case study
- Fixed turbine type
- No economic optimization
- No cable routing optimization
- No bathymetry constraints
- Simplified wind climate assumptions

Discuss how these assumptions may influence the conclusions and how future work could address these limitations.

8 Conclusion

9 Future work

9.1 Computational Efficiency

One of the main challenges during this study was the computational efficiency, which resulted in several modeling decisions being made based on runtime limitations and computational cost. With access to a more powerful computer platform and higher CPU performance, more advanced and computationally demanding simulations could have been investigated. This study only investigates the Horns Rev 1 offshore wind farm using a simplified AEP estimation approach. To extend the framework to more complex offshore wind farms with irregular polygon boundaries and obtain more realistic AEP estimations, more advanced models and significantly higher computational efficiency would be required.

9.2 LCOE

In this study, the only objective function considered is the maximization of AEP. However, when planning an offshore wind farm, many additional factors influence the final wind farm design. One of the most important factors is the economic aspect. The Levelized Cost of Energy (LCOE) is a method used to combine all major cost components, including wind turbines, monopiles, cables, substations, operation and maintenance, and other installation-related costs. These components play a crucial role in the total cost of establishing an offshore wind farm. In particular, the water depth at the turbine locations strongly affects the monopile foundation costs, which typically account for approximately 13–15% of the total offshore wind farm cost. In addition, the cable layout must be carefully planned due to the high installation and material costs associated with offshore array cables. Including LCOE as part of the optimization objective would therefore provide a more realistic optimization framework, where not only the AEP is considered, but also the trade-off between energy production, economics, and potentially CO_2 emissions.

9.3 Wake Models

As described in **??**, the wake model comparison showed noticeable differences between the NOJ wake model and the more advanced Bastankhah Gaussian wake model. Using a more advanced wake loss model would improve the physical accuracy of the simulations and provide a more realistic representation of wake behavior and turbulence intensity within the wind farm.w More advanced wake models generally capture wake recovery, turbulence effects, and wake interactions more accurately, especially for large offshore wind farms. However, these models also increase the computational cost significantly, which again highlights the trade-off between model accuracy and computational efficiency.

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A Appendix